Advanced Counting

① Recurrence Relation

⇒ A recurrence relation for the sequence $\{a_n\}$ is an equation that expresses a_n in terms of one or more of the previous terms of the sequence, namely $a_0, a_1, \ldots, a_{n-1}$ for all integers n with $n \geq n_0$ where n_0 is non-negative integer.

eg: Let $\{a_n\}$ be a sequence that satisfies the recurrence relation $a_n = a_{n-1} - a_{n-2}$ for $n = 2, 3, 4, \ldots$ and suppose that $a_0 = 3$ and $a_1 = 5$. What are a_2 and a_3?

Soln Given, $a_n = a_{n-1} - a_{n-2}$

$$a_2 = a_{2-1} - a_{2-2} = a_1 - a_0 = 5 - 3 = 2.$$

$$a_3 = a_{3-1} - a_{3-2} = a_2 - a_1 = 2 - 5 = -3.$$

⊛ Find the recurrence relation of the sequence 2, 5, 11, 23, 47.

Soln

$$a_1 = 2$$

$$a_2 = 5 = 2\times2+1 = 2\times a_1 + 1$$

$$a_3 = 11 = 2\times5+1 = 2\times a_2 + 1$$

$$a_4 = 23 = 2\times11+1 = 2\times a_3 + 1$$

$$a_5 = 47 = 2\times23+1 = 2\times a_4 + 1.$$

$$\vdots \qquad \vdots \qquad \vdots$$

$$a_n = 2\times a_{n-1} + 1$$

Hence, the recurrence relation is $a_n = 2a_{n-1} + 1$ with initial conditions $a_1 = 2$, $n \geq 2$.

CW ⊛ Find the first five terms of the sequence defined by $a_n = 6a_{n-1}$, $a_0 = 2$.

Recurrence relation for fibonacci numbers.

$f_n = f_{n-1} + f_{n-2}$ $n \geq 3$ with the initial conditions $f_1 = 1$ and $f_2 = 1$.

Solving Linear Recurrence Relations.

Linear homogeneous recurrence relation

$\Rightarrow$ A linear homogeneous recurrence relation of degree k with constant coefficient is a recurrence relation of the form

$$a_n = c_1 a_{n-1} + c_2 a_{n-2} + \cdots + c_k a_{n-k}$$

Where $c_1, c_2, \ldots, c_k$ are real numbers and $c_k \neq 0$.

Example 1.
The recurrence relation $p_n = (1.11) p_{n-1}$ is a linear homogeneous recurrence relation of degree 1.

Example 2: The recurrence relation $f_n = f_{n-1} + f_{n-2}$ is a linear homogeneous recurrence relation of degree 2.

Theorem 1: Let c_1 and c_2 be real numbers. Suppose that $r^2 - c_1 r - c_2 = 0$ has two distinct roots r_1 and r_2. Then the sequence $\{a_n\}$ is a solution of the recurrence relation $a_n = c_1 a_{n-1} + c_2 a_{n-2}$ if and only if $a_n = \alpha_1 r_1^n + \alpha_2 r_2^n$ for $n = 0, 1, 2, \ldots$ where α_1 and α_2 are constants.

1. What is the solution of the recurrence relation
$$a_n = a_{n-1} + 2a_{n-2} \quad \text{with } a_0 = 2 \text{ and } a_1 = 7 ?$$

Sol^n Theorem 1 can be used to solve this recurrence relation.
The characteristic equation of the recurrence relation is,
$$r^2 - c_1 r - c_2 = 0.$$
i.e. $c_1 = 1$ and $c_2 = 2$.
So, $r^2 - c_1 r - c_2 = 0$
or, $r^2 - r - 2 = 0 \Rightarrow r^2 - 2r + r - 2 = 0 \Rightarrow r(r-2) + 1(r-2) = 0$
It's roots are $r = 2$ and $r = -1$

Hence, the sequence $\{a_n\}$ is solution to recurrence relation if and only if
$$a_n = \alpha_1 r_1^n + \alpha_2 r_2^n$$

$a_n = \alpha_1 2^n + \alpha_2(-1)^n$. for constants α_1 and α_2.

From the initial condition it follows that

$a_0 = \alpha_1 2^0 + \alpha_2(-1)^0$

or, $2 = \alpha_1 + \alpha_2$

$a_1 = \alpha_1 2^1 + \alpha_2(-1)^1$

or, $7 = 2\alpha_1 - \alpha_2$

Solving these two equations we get,

$\alpha_1 = 3$ and $\alpha_2 = -1$

Hence, the solution is

$a_n = 3 \cdot 2^n - (-1)^n$. Ans

(*) Find the explicit formula for the fibonacci numbers.

Sol^n The sequence of fibonacci numbers satisfies the recurrence relation $f_n = f_{n-1} + f_{n-2}$ with initial condition $f_0 = 1$ and $f_1 = 1$.

The roots of the characteristics equations are

$r^2 - c_1 r - c_2 = 0$

or, $r^2 - r - 1 = 0$

Solving we get,

$r_1 = (1+\sqrt{5})/2$ and $r_2 = (1-\sqrt{5})/2$.

From theorem 1, it follows that the fibonacci numbers are given by,

$$f_n = \alpha_1 \left(\frac{1+\sqrt{5}}{2}\right)^n + \alpha_2 \left(\frac{1-\sqrt{5}}{2}\right)^n$$

for some constants α_1 and α_2.

$f_0 = \alpha_1 + \alpha_2$

$$f_1 = \alpha_1 \left(\frac{1+\sqrt{5}}{2}\right) + \alpha_2 \left(\frac{1-\sqrt{5}}{2}\right)$$

Solving these, we get

$\alpha_1 = 1/\sqrt{5}$, $\alpha_2 = -1/\sqrt{5}$

Consequently, the fibonacci sequence is given by,

$$f_n = \frac{1}{\sqrt{5}}\left(\frac{1+\sqrt{5}}{2}\right)^n - \frac{1}{\sqrt{5}}\left(\frac{1-\sqrt{5}}{2}\right)^n.$$

Ans

~~Theorem 2:~~

HW

(*) Solve the recurrence relation

$a_n = 6a_{n-1} - 8a_{n-2}$ for $n \geq 2$, $a_0 = 4$, $a_1 = 10$

Soln $a_n = 1.4^n + 3.2^n$.

(*) What is the solution of the recurrence relation

$a_n = a_{n-1} + 2a_{n-2}$ with $a_0 = 2$ and $a_1 = 7$?

Soln $a_n = 3.2^n - 1.(-1)^n$.

Theorem 2

Let c_1 and c_2 be real numbers with $c_2 \neq 0$. suppose that $r^2 - c_1 r - c_2 = 0$ has only one root r_0. A sequence $\{a_n\}$ is a solution of the recurrence relation $a_n = c_1 a_{n-1} + c_2 a_{n-2}$ if and only if $a_n = \alpha_1 r_0^n + \alpha_2 n r_0^n$ for $n = 0, 1, 2, \ldots$, where α_1 and α_2 are constants.

(*) What is the solution of the recurrence relation

$a_n = 6a_{n-1} - 9a_{n-2}$, with initial conditions $a_0 = 1$ and $a_1 = 6$?

Soln

Comparing $a_n = 6a_{n-1} - 9a_{n-2}$ with $a_n = c_1 a_{n-1} + c_2 a_{n-2}$

we get $c_1 = 6$, $c_2 = -9$.

The characteristic eqn is, $r^2 - c_1 r - c_2 = 0$

or, $r^2 - 6r + 9 = 0$

or, $r^2 - 3r - 3r + 9 = 0$

or, $r(r-3) - 3(r-3) = 0$

So, $r = 3$

Hence, the solution to recurrence relation is
$a_n = \alpha_1 3^n + \alpha_2 \cdot n \cdot 3^n$ for some constants α_1 and α_2.

$a_0 = \alpha_1 3^0 + \alpha_2 \cdot 0 \cdot 3^0$
or, $1 = \alpha_1$
$a_1 = \alpha_1 \cdot 3^1 + \alpha_2 \cdot 1 \cdot 3^1$
or, $6 = 3\alpha_1 + 3\alpha_2$
Solving these, we get,
$\alpha_1 = 1$ and $\alpha_2 = 1$.
So, the solution to recurrence relation is,
$a_n = 3^n + n \cdot 3^n$ Ans

H.W

- Solve the recurrence relation $a_n = 2a_{n-1} - a_{n-2}$ for $n \geq 2$, $a_0 = 4$, $a_1 = 1$.

  Ans: $a_n = 4 \cdot 1^n + (-3) \cdot n \cdot 1^n$.

- Solve the recurrence relation $a_n = 2a_{n-1} - a_{n-2}$ for $n \geq 2$ with $a_0 = 3$ and $a_1 = 6$.

  Ans: $3 \cdot 1^n + 3 \cdot n \cdot 1^n$. ~~Ans~~

Theorem 3:
Let $c_1, c_2, \ldots, c_k$ be real numbers. Suppose that the characteristic equation
$r^k - c_1 r^{k-1} - \cdots - c_k = 0$ has 'k' distinct roots $r_1, r_2, \ldots, r_k$. Then a sequence $\{a_n\}$ is a solution of the recurrence relation
$a_n = c_1 a_{n-1} + c_2 a_{n-2} + \cdots + c_k a_{n-k}$ if and only if
$a_n = \alpha_1 r_1^n + \alpha_2 r_2^n + \cdots + \alpha_k r_k^n$
for $n = 0, 1, 2, \ldots$ where $\alpha_1, \alpha_2, \ldots, \alpha_k$ are constants.

(*) Find the solution of the recurrence relation

$$a_n = 6a_{n-1} - 11a_{n-2} + 6a_{n-3}$$

with the initial conditions $a_0 = 2, a_1 = 5$ and $a_2 = 15$.

Soln The characteristics equation of the polynomial is,

$$r^3 - 6r^2 + 11r - 6 = 0$$

$$(r-1)(r-2)(r-3) = 0.$$

So, we get $r = 1, r = 2$ and $r = 3$.

So, the solution of the recurrence relation is in the form

$$a_n = \alpha_1 . 1^n + \alpha_2 . 2^n + \alpha_3 . 3^n$$

$$a_0 = 2 = \alpha_1 + \alpha_2 + \alpha_3$$

$$a_1 = 5 = \alpha_1 + 2\alpha_2 + 3\alpha_3$$

$$a_2 = 15 = \alpha_1 + 4\alpha_2 + 9\alpha_3$$

Solving these equations, we get

$$\alpha_1 = 1; \alpha_2 = -1 \text{ and } \alpha_3 = 2.$$

Hence, the solution of the recurrence relation is,

$$a_n = 1.1^n - 1.2^n + 2.3^n.$$ Ans

H.W (*) Solve the recurrence relation $a_n = 2a_{n-1} + a_{n-2} - 2a_{n-3}$ for $n \geq 3$, $a_0 = 3, a_1 = 6$ and $a_2 = 9$.

Ans: $a_n = (3/2) 1^n - (1/2)(-1)^n + 2.2^n$.

(*) Find the solution of the recurrence relation $a_n = 6a_{n-1} + 11a_{n-2} + 6a_{n-3}$ for $n > 3$, $a_0 = 2, a_1 = 5$ and $a_2 = 5$.

Ans: $a_n = -4.1^n - 3.3^n + 9.2^n$.

Theorem 4:

Let $c_1, c_2, \ldots, c_k$ are real numbers. Suppose that the characterstic equation $r^k - c_1 r^{k-1} - \cdots - c_k = 0$ has 't' distinct roots $r_1, r_2, \ldots, r_t$ with multiplicities $m_1, m_2, \ldots, m_t$, respectively. so, that $m_i \geq 1$ for $i = 1, 2, \ldots, t$ and $m_1 + m_2 + \cdots + m_t = k$. Then a sequence $\{a_n\}$ is a solution of the recurrence relation

$$a_n = c_1 a_{n-1} + c_2 a_{n-2} + \cdots + c_k a_{n-k} \text{ if and only if}$$

$$a_n = (\alpha_{1,0} + \alpha_{1,1} n + \cdots + \alpha_{1,m_1-1} n^{m_1-1}) r_1^n$$
$$+ (\alpha_{2,0} + \alpha_{2,1} n + \cdots + \alpha_{2,m_2-1} n^{m_2-1}) r_2^n$$
$$+ \cdots$$
$$+ (\alpha_{t,0} + \alpha_{t,1} n + \cdots + \alpha_{t,m_t-1} n^{m_t-1}) r_t^n$$

for $n = 0, 1, 2, \ldots$ where $\alpha_{i,j}$ are constants for $1 \leq i \leq t$ and $0 \leq j \leq m_i - 1$.

Find the solution of the recurrence relation

$$a_n = -3a_{n-1} - 3a_{n-2} - a_{n-3}$$

with initial conditions $a_0 = 1, a_1 = -2$ and $a_2 = -1$.

Soln

The characterstic equation of the recurrence relation is,

$$r^3 + 3r^2 + 3r + 1 = 0$$

Solving these equations we get, ~~$r = 1$~~. $(r+1)^3 = 0$ i.e $r = -1, -1, -1$

By theorem 4, the solution of recurrence relation is,

$$a_n = \alpha_{1,0}(-1)^n + \alpha_{1,1}(-1)^n \cdot n + \alpha_{1,2} n^2 (-1)^n.$$

using initial conditions,

$$a_0 = 1 = \alpha_{1,0}$$

$$a_1 = -2 = -\alpha_{1,0} - \alpha_{1,1} - \alpha_{1,2}.$$

$$a_2 = -1 = \alpha_{1,0} + 2\alpha_{1,1} + 4\alpha_{1,2}$$

Solving these equations, we get,

$\alpha_{1,0} = 1$

$\alpha_{1,1} = 3$

$\alpha_{1,2} = -2.$

Hence, $a_n = (1+3n-2n^2)(-1)^n$. Ans

Linear non-homogeneous recurrence relation with constant coefficients

→ In linear non-homogeneous recurrence relation, the recurrence relation is of the form

$$a_n = c_1 a_{n-1} + c_2 a_{n-2} + \cdots + c_k a_{n-k} + F(n)$$

where $c_1, c_2, \ldots, c_k$ are real numbers and $F(n)$ is a function depending on 'n'.

→ The recurrence relation preceeding $F(n)$ is called associated homogeneous recurrence relation.

eg: $a_n = 7a_{n-1} + 3a_{n-2} + 6n$ is linear non-homogeneous recurrence relation with constant coefficients.

Theorem 5: If $\{a_n^{(p)}\}$ is a particular solution of the non-homogeneous linear recurrence relation with constant coefficients

$$a_n = c_1 a_{n-1} + c_2 a_{n-2} + \cdots + c_k a_{n-k} + F(n).$$

then every solution is of the form $\{a_n^{(p)} + a_n^{(h)}\}$ where $\{a_n^{(h)}\}$ is a solution of the associated homogeneous recurrence relation.

$$a_n = c_1 a_{n-1} + c_2 a_{n-2} + \cdots + c_k a_{n-k}.$$

(*) Find all solution of the recurrence relation $a_n = 3a_{n-1} + 2n$. What is the solution with $a_1 = 3$?

Soln To solve this linear non-homogeneous recurrence relation with constant coefficients, we need to solve its associated linear homogeneous equation and to find particular solution for the given non-homogeneous equation.

The associated linear homogeneous equation is $a_n = 3a_{n-1}$ — eq^n (1)

From theorem 3, the characteristic equation is,

$$r - 3 = 0 \Rightarrow r = 3.$$

So, the solution of eq^n (1) is

$$a_n = \alpha_1 . r_1^n = \alpha_1 . 3^n \quad \text{— eq}^n \text{ (ii)}$$

To find particular solution, we write

$$F(n) = 2n \quad \text{— eq}^n \text{ (iii)}$$

The trial solution of eq^n (iii) is

~~$a(p) = c.2n$~~ $a_n(p) = cn + d$ where c and d are constants.

Substituting terms of this sequence into recurrence relation, we have

$$a_n = 3a_{n-1} + 2n$$ [put $a_n(p) =$ ~~c.2n~~ $cn+d$ in both sides]

$c.2n = 3.c.2(n-1) + 2n$

or, $c.2n = 3.c.(2n-2) + 2n$

or, $c.2n = 6cn - 6c + 2n$

$$cn + d = 3(c(n-1) + d) + 2n$$

Simplifying and combining like terms gives.

$$cn + d = 3(cn - c + d) + 2n$$

or $cn + d = 3cn - 3c + 3d + 2n$

or $cn + d - 3cn + 3c - 3d - 2n = 0$

or $-2cn - 2d + 3c - 2d - 2n = 0$

or $(2 + 2c)n + (2d - 3c) = 0$

It follows that $cn+d$ is a solution iff $2 + 2c = 0$ and $2d - 3c = 0$

This shows that $cn+d$ is a solution iff $c = -1$ and $d = -3/2$.

Consequently, $a_n(p) = -n - \frac{3}{2}$ [$\because a_n(p) = cn+d$]

By theorem 5, all solutions are of the form
$a_n = a_n^{(p)} + a_n^{(h)} = -n - \frac{3}{2} + \alpha \cdot 3^n$, where α is constant.

To find solution with $a_1 = 3$ let $n=1$.
$a_1 = 3 = -1 - \frac{3}{2} + \alpha \cdot 3^1$
or, $\alpha = 11/6$.
So, the final solution is,
$$a_n = -n - \frac{3}{2} + (11/6) \cdot 3^n. \text{ Ans}$$

(*) Find all the solutions of the recurrence relation
$$a_n = 5a_{n-1} - 6a_{n-2} + 7^n;$$
Soln This is linear non-homogeneous recurrence relation. The solution of its associated homogeneous recurrence relation is,
$$a_n = 5a_{n-1} - 6a_{n-2}$$
are, $a_n^{(h)} = \alpha_1 \cdot 3^n + \alpha_2 \cdot 2^n$ where α_1 & α_2 are constants.
$F(n) = 7^n$, so, a reasonable trial solution is $a_n^{(p)} = c \cdot 7^n$
Where c is constant.
Substituting the terms in the recurrence relation.
$$c \cdot 7^n = 5 \cdot c \cdot 7^{n-1} - 6 \cdot c \cdot 7^{n-2} + 7^n$$
Factoring out 7^{n-2}, the equation becomes
$$49c = 35c - 6c + 49 \Rightarrow 20c = 49 \Rightarrow c = 49/20$$
Hence, $a_n^{(p)} = (49/20) \cdot 7^n$.

So, By theorem 5, the solution is,
$$a_n = \alpha_1 \cdot 3^n + \alpha_2 \cdot 2^n + (49/20) \cdot 7^n. \text{ Ans}$$

H.W

(*) Find all the solution of the recurrence relation $a_n = 2a_{n-1} + 3^n$ and a solution with initial condition $a_1 = 5$.

let $a_n^{(p)} = c \cdot 3^n$.

① Find all the solutions of the recurrence relation $a_n = 4a_{n-1} + n^2$. Also find the solution of the relation with initial condition $a_1 = 1$. ?

Let. $a_n(p) = an^2 + bn + c$.

where a, b, c are constants.

Theorem: 6. Suppose that $\{a_n\}$ satisfies the linear nonhomogeneous recurrence relation

$$a_n = c_1 a_{n-1} + c_2 a_{n-2} + \cdots + c_k a_{n-k} + F(n)$$ where $c_1, c_2, \ldots, c_k$ are real numbers and

$$F(n) = (b_t n^t + b_{t-1} n^{t-1} + \cdots + b_1 n + b_0) s^n$$

where $b_0, b_1, \ldots, b_t$ and s are real numbers. Where s is not a root of the characteristic equation of the associated linear homogeneous recurrence relation, there is particular solution of the form.

$$(p_t n^t + p_{t-1} n^{t-1} + \cdots + p_1 n + p_0) s^n$$

Where s is root of the characteristic equation and it's multiplicity is m, there is particular solution of the form

$$n^m (p_t n^t + p_{t-1} n^{t-1} + \cdots + p_1 n + p_0) s^n.$$

(*) What form does a particular solution of the linear non-homogeneous recurrence relation $a_n = 6a_{n-1} - 9a_{n-2} + F(n)$ have when $F(n) = 3^n$, $F(n) = n \cdot 3^n$, $F(n) = n^2 \cdot 2^n$ and $F(n) = (n^2+1) \cdot 3^n$?

Soln The associated linear homogeneous recurrence relation is,

$$a_n = 6a_{n-1} - 9a_{n-2}$$

The characteristic equation is

$$r^2 - 6r + 9 = 0 = (r-3)^2 = 0,$$ has a single root $r = 3$ with multiplicity 2.

To apply theorem 6, with $F(n)$ of the form $p(n) \cdot s^n$ where $p(n)$ is polynomial and s is constant. We need to ask whether s is root of this characteristic equation.

Because $s=3$ is a root with multiplicity $m=2$ but $s=2$ is not a root, Theorem 6 tells us that a particular solution has the form $p_0 n^2 3^n$, the form $n^2(p_1 n+p_0)3^n$ if $F(n)=n\cdot 3^n$, the form $(p_2 n^2+p_1 n+p_0)\cdot 2^n$ if $F(n)=n^2 2^n$ and the form $n^2(p_2 n^2+p_1 n+p_0)3^n$ if $F(n)=(n^2+1)\cdot 3^n$
Ans

Recurrence Relation Applications.

→ One of the important application areas of recurrence relation is analysis of divide and conquer algorithms.

Divide and conquer algorithm

→ The divide and conquer algorithm, divides the given problem into smaller sub-problems and the solution of smaller sub-problems are combined together to get the solution of the original problem.

→ The divide and conquer recurrence relation is

$$f(n) = af(n/b) + g(n)$$

In this, the problem of size 'n' is divided into problem of size n/b and $g(n)$ is the operations required to conquer the solutions.

Eg: ① Fibonacci numbers.

$f_n = f_{n-1} + f_{n-2}$ $f_0 = 1, f_1 = 1$ and $n \geq 3$.

② merge sort.

$M(n) = 2M(n/2) + O(1)$

③ Binary search.

$f(n) = f(n/2) + 2.$